

KALPANA KANUKA

DATA ENGINEER

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SUMMARY

Data Engineer with 3+ years of experience designing and optimizing cloud-based data pipelines, streaming frameworks, and analytics solutions for healthcare and technology sectors. Proficient in implementing scalable ETL systems, automating workflows, and deploying AI-driven models to enhance efficiency, data quality, and real-world decision-making.

EDUCATION

Master of Science in Computer Science | University of Bridgeport, Bridgeport, CT

May 2025

Bachelor of Technology in Computer Science | Narsimha Reddy Engineering College, India

SKILLS

Programming & Data Engineering Tools: Python, SQL, R, PySpark, Bash/Shell Scripting, Data Modeling, Data Warehousing, Git, Jenkins, Docker, Kubernetes, CI/CD Pipelines, Agile Tools

Machine Learning & AI: Supervised, Unsupervised, Reinforcement Learning, NLP, LLMs, Transformers, Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Keras, Hugging Face, LangChain, MLflow, Vector Databases

ETL & Data Integration: Azure Data Factory (ADF), Azure Synapse Analytics, Azure Databricks, AWS Glue, DBT (Data BuildTool), Informatica, Talend, Fivetran, Stitch Big Data & Distributed Systems: Apache Spark, Kafka, Hadoop HDFS, Apache Airflow, Flink, Delta Lake, Databricks Lakehouse

Cloud Platforms: **AWS** (S3, Redshift, EMR, SageMaker, Lambda), **Azure** (Synapse, Databricks, Cosmos DB), **Google Cloud** (BigQuery, Dataflow, Vertex AI, Pub/Sub)

Databases & Storage: PostgreSQL, MySQL, Microsoft SQL Server, Cassandra, Redis, Snowflake, Databricks Delta Tables

Visualization & BI Tools: Power BI, Tableau, Looker, KPI Dashboards, Executive Reporting, Plotly, Altair

Collaboration & Microsoft Tools: Office Suite (Excel, Word, PowerPoint, Access, Outlook), SharePoint, Teams

WORK EXPERIENCE

Data Engineer | CVS Health, TX, USA

Jan 2025 – Present

- Design and maintain scalable data pipelines with Azure Data Factory, Databricks, and Synapse Analytics, processing 5M+ claims and provider records daily with HIPAA-compliant practices.
- Implement streaming pipelines using Kafka and Spark Structured Streaming, enabling near real-time monitoring of patient outcomes and reducing reporting lag by 40%.
- Optimize ETL workflows with PySpark and SQL, cutting daily data refresh times by 30% and improving reliability of downstream analytics.
- Collaborate with AI teams to operationalize predictive models for patient risk stratification using MLflow and Databricks, improving accuracy of readmission predictions by 15%.
- Engineer feature pipelines for NLP models that process unstructured physician notes, enabling faster insights from 100K+ clinical documents per week.
- Integrate vector databases (Pinecone, FAISS) into retrieval pipelines, supporting clinical staff with retrieval-augmented search across 10TB+ of EHR and claims data.
- Build MLOps workflows with CI/CD (Git, Jenkins, Docker, Kubernetes) to automate AI model deployment, reducing release cycle times by 35%.
- Deliver Power BI executive dashboards combining AI-driven predictions with KPI metrics, improving decision-making in care management and reducing readmission rates by 12%.

Jr. Data Engineer | Hexaware, INDIA

Mar 2021 – Jul 2023

- Migrated 1TB+ of on-premises data warehouses into AWS Redshift using Glue and DBT, which reduced storage costs by 22% and improved query response times for analysts across finance and operations.
- Automated ingestion from APIs, flat files, and IoT sources by building Airflow DAGs, cutting manual intervention by 40% and reducing pipeline failure rates by 35% to ensure SLA compliance.
- Engineered PySpark feature pipelines that delivered curated datasets for fraud detection and churn models, increasing data availability by 25% and enabling faster delivery of model insights to stakeholders.
- Designed Delta Lake-based workflows to improve training data freshness for machine learning models, which shortened retraining cycles by 20% and enhanced prediction accuracy for customer analytics.
- Processed raw application and customer logs into structured datasets using Spark NLP and Python, enabling sentiment analysis and accelerating reporting turnaround for business teams by 15%.
- Collaborated with cross-functional teams including data scientists, BI developers, and product managers to align data engineering solutions with business goals, ensuring timely delivery of analytics initiatives.